

A Review on Perovskite-Type LaFeO_3 Based Electrodes for CO_2 Reduction in Solid Oxide Electrolysis Cells: Current Understanding of Structure–Functional Property Relationships

Mykhailo Pidburnyi, Bryan Zanca, Claude Coppex, Santiago Jimenez-Villegas, and Venkataraman Thangadurai*



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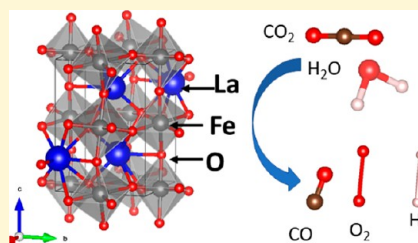


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ABSTRACT: Mixed ionic and electronic conductors (MIECs) are being studied extensively as a potential replacement for cermet catalysts in solid oxide cells (SOCs)—collectively named for solid oxide fuel cells and solid oxide electrolysis cells—due to their high activity, great stability, and low cost. Perovskite-type LaFeO_3 -based catalysts are one of the most promising electrodes for SOCs. This review paper provides an update on the chemical composition–crystal structure–physical property relationships and an understanding of the electrochemical reaction mechanisms in the perovskite-type structure LaFeO_3 electrodes. Mechanisms for electrochemical CO_2 reduction on the perovskite structure surface and oxygen diffusion are discussed. Furthermore, the electrochemical performance enhancement strategies such as composite electrodes and nanoparticles ex-solution are briefly presented together with industrialization issues, e.g., gas impurities and stability of the interconnecting materials.



1. INTRODUCTION

High temperature solid oxide electrolysis cells (SOECs) are a promising method for syngas production and organic chemicals synthesis. The main advantage of the SOEC is that fossil fuels burning byproducts can be used as a feeding gas, which can reduce greenhouse gas emissions. The SOEC can be used in the fuel cell mode to produce electricity with a high efficiency (Figure 1). Many different materials are currently being developed to be used as electrode catalysts for SOECs. There are some requirements such as high catalytic activity for redox reactions, stability in operating environments, and chemical stability with cell components. The most obvious candidate is a widely used composite Ni–Y-doped ZrO_2 (Ni-YSZ) cermet electrode for solid oxide fuel cells (SOFCs). However, in metal-ceramic electrodes the electrochemical reactions only occur at the so-called triple-phase boundary (TPB) where the ionic conducting phase (YSZ), electronic conducting phase (Ni), and gas phase meet. Hence, the cell performance is limited by the availability of the TPB length.¹ Another issue is that Ni particles oxidize when water and water/ CO_2 coelectrolysis is taking place, which leads to NiO formation and a loss of electronic conductivity.² Moreover, Ni-YSZ cermets are prone to carbon formation, sulfur poisoning, and Ni segregation.^{3,4}

Mixed ionic and electronic conductors (MIECs) are being developed to replace the Ni-YSZ electrode. These MIEC materials demonstrate high catalytic activity, redox stability, and sulfur tolerance compared to Ni-YSZ.^{5,6} Furthermore, several MIECs have demonstrated high stability in CO_2/CO at

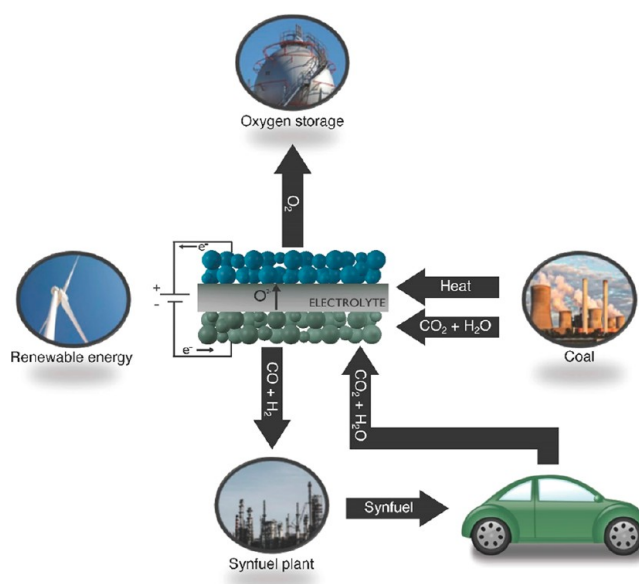


Figure 1. Solid oxide fuel cell (SOFC)/solid oxide electrolysis cell (SOEC) integration into renewable and nonrenewable energy recourses and its application.

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an elevated temperature.^{7–9} in which the entire surface is electrochemically active leading to a significant performance improvement, unlike cermet electrodes, due to MIEC–gas interface activity (dual-phase-boundary, DPB). Fluorite-type $\text{Ce}_{0.9}\text{Gd}_{0.1}\text{O}_{2-\delta}$, perovskite-type LaFeO_3 , $\text{La}_{0.75}\text{Sr}_{0.25}\text{Cr}_{0.5}\text{Mn}_{0.5}\text{O}_{3-\delta}$, double perovskite-type $\text{Sr}_2\text{MgMoO}_6$, and Nd_2WO_6 oxides are reported as potential electrodes for SOCs.^{10–13} Among these families of MIECs, doped lanthanum ferrites were recently developed as SOEC cathodes due to their good catalytic activity, electronic conductivity, and oxygen mobility.^{14–16} Since perovskite-type structure oxides are currently being widely explored, this article is aimed to provide an overview of LaFeO_3 electrodes for SOECs.

2. STRUCTURE PROPERTIES OF LaFeO_3 AND ITS RELATED MATERIALS

The naturally occurring perovskite mineral, CaTiO_3 , has simple origins, but the complexity built around the synthetic perovskite-like structure with a general chemical formula of ABX_3 (A = mono-, di-, and or trivalent cations; B = tri-, tetra-, and pentavalent cations; X = mono-, di-, and trivalent anions). The ideal cubic perovskite crystal structure is built up of AX_3 close packed layers with 1/4 of the octahedral holes filled with B-site cations. The A-site cation is usually large (~ 1.4 Å) and coordinated by 12 anions, and the B-site is small (~ 0.6 Å) and octahedrally coordinated by anions. The B-site octahedra form a framework of corner sharing octahedra, and depending on the size of the B-site cation different interstices spaces are created that can accommodate different ionic radius A-site cations in the perovskite framework.

The crystal structure of a primitive cubic perovskite $\text{ABO}_3 \cong \text{SrTiO}_3$ is shown in Figure 2. The ideal cubic perovskite

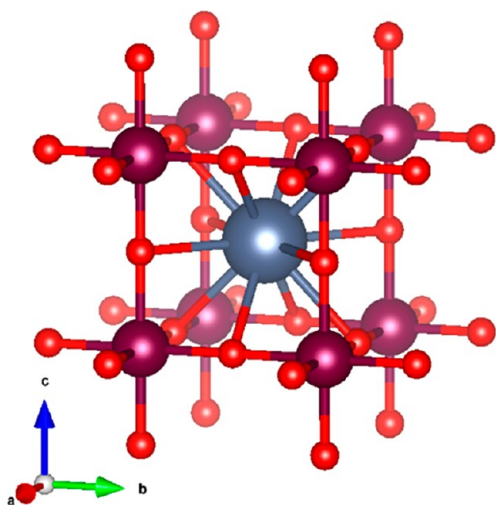


Figure 2. Primitive cubic perovskite (ABO_3) unit cell of SrTiO_3 (Sr: blue, Ti: purple; O: red).

structure is not very common due to the lack of cations with the ideal ionic radii on their ideal crystallographic sites. When the A-site cation is smaller than the Sr^{2+} ion, a phenomenon known as octahedral tilting occurs. This leads to structural modifications that produce interesting electrical and magnetic properties. As the perovskite structure diverts from the ideal cubic form, octahedral tilts act to stabilize the structure, and the degree or the nature of tilting has been explained using a

Glazer Notation.¹⁷ The distortions in perovskites can usually be explained using three common principles which include size, a change in oxidation states, and Jahn–Teller effects.^{18,19}

In order to determine the stability of possible perovskite materials, Goldschmidt developed a formula that uses the geometric structural identity of the perovskite structure and the ionic radii of the A- and B-site cations (r_A and r_B) and anion (r_O).²⁰ This Goldschmidt tolerance factor (t) is defined as

$$t = \frac{(r_A + r_O)}{\sqrt{2}(r_B + r_O)} \quad (1)$$

The Goldschmidt tolerance factor when equal to 1 represents an ideal cubic perovskite (e.g., SrTiO_3). If the t number drops below 0.9, the structure shifts toward either an orthorhombic or a rhombohedral distortion caused by different octahedral tilting (e.g., CaTiO_3). When the tolerance factor is greater than 1, the perovskite structure typically shifts toward a tetragonal or hexagonal distortion (e.g., BaTiO_3 and BaNiO_3).

LaFeO_3 Structural Characteristics. LaFeO_3 has been proposed as a promising cathode for the intermediate temperature (IT) SOFCs and for the electrochemical reduction of CO_2 in SOECs.^{21,22} This material displays MIEC properties that can be controlled both intrinsically and extrinsically.^{23,24} Therefore, a good understanding of the crystal structure is important to understand the nature of defects that control the functional properties of the LaFeO_3 material. The structure of undoped LaFeO_3 adopts an orthorhombic (space group $Pbnm$ (No. 62)) perovskite arrangement at ambient temperatures and up to ~ 957 °C. The crystal structure of LaFeO_3 is shown in Figure 3.²¹ The unit cell at room temperature has four structural units ($Z = 4$) and edge lengths of $a = 5.55$ Å, $b = 5.56$ Å, and $c = 7.86$ Å.²¹ Tables 1 and 2 list the structural parameters of LaFeO_3 . There are two oxygen inequivalent atomic positions; the first oxygen (O1) is tetrahedrally coordinated by two lanthanum cations and two iron cations and sits within the layers of lanthanum, and the second oxygen (O2) is coordinated by three lanthanide cations and two iron cations and rests between the layers of iron. The Goldschmidt tolerance factor for LaFeO_3 is 0.9543, which indicates that this material should crystallize in the cubic perovskite structure. This deviation from the cubic form is commonly caused by the A-site cation being too small to fit in the interstices of the B-site cations.¹⁸ The structure is stabilized by neighboring octahedra tilting in opposite directions; this can be described using glazer notation as $a^+b^-b^-$ or $a^+a^-a^-$ depending on the space group setting ($Pbmn$ vs $Pnma$, respectively).¹⁸ The two tilting rotations are opposite in direction (out-of-phase) but equal in magnitude. The unit cell can be related to a pseudocubic perovskite lattice by $a \approx b \approx 2^{1/2}a_p$ and $c \approx 2a_p$. The expansion of the unit cell is related to the degree of octahedral tilting observed. The nature of the octahedral tilt observed in LaFeO_3 has been discussed by Woodward to be caused by a strong covalent bond interaction between lanthanum and oxygen, and the structure distorts in order to maximize the degree of orbital overlap.¹⁹ This displacement of La^{3+} cations is projected along the ab -plane.

LaFeO_3 undergoes a first order structural phase transition from orthorhombic to rhombohedral between ~ 960 and ~ 1005 °C (Figure 4).²² The nature of this nonlinear thermal expansion can be explained by the octahedral tilting of the B-

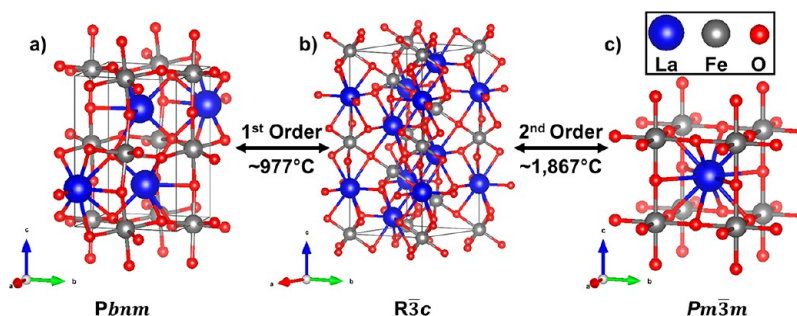


Figure 3. Perovskite structure of LaFeO_3 adopting three distinct phases, (a) a low temperature orthorhombic phase ($Pbnm$), (b) a high temperature rhombohedral phase ($R\bar{3}c$), and (c) a high temperature cubic phase ($Pm\bar{3}m$).

Table 1. Crystallographic Data for LaFeO_3 in Space Group $Pbnm$

element	Wyckoff Symbol	x	y	z	occupancy
lanthanum (La)	4c	−0.00614	0.02840	1/4	1
iron (Fe)	4b	0	1/2	0	1
oxygen (O1)	4c	0.0731	0.4875	1/4	1
oxygen (O2)	8d	−0.2809	0.2815	0.0394	1

Table 2. Crystallographic Data for LaFeO_3 in Space Group $R\bar{3}c$

element	Wyckoff symbol	x	y	z	occupancy
lanthanum (La)	6a	0	0	1/4	1
iron (Fe)	6b	0	0	0	1
oxygen (O)	18e	0.2200	1/3	0.0833	1

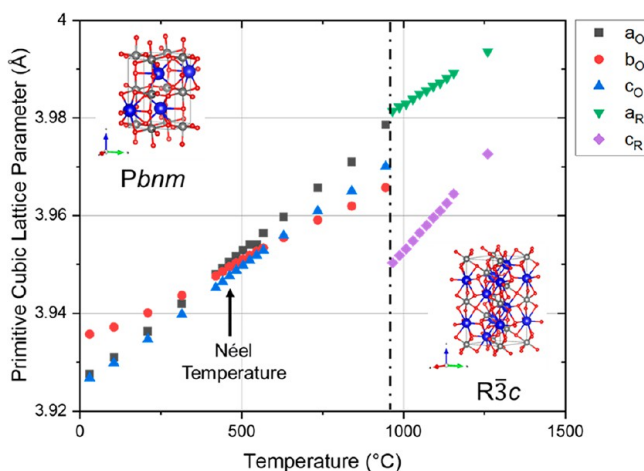


Figure 4. Nonlinear thermal expansion of the lattice parameters that can be observed in the primitive cubic lattice parameters between 30 and 1200 °C.³⁴ The dotted line marks the phase transition from orthorhombic ($Pbnm$) to rhombohedral ($R\bar{3}c$) at 966 °C. The Néel transition is marked at 462 °C.

site cations, which can be further correlated to the observed magnetic properties, which has a Néel transition temperature at $T_N = 462$ °C.²² It can be seen from Figure 4 that the a and b axes expand in a linear fashion; the c axis expands in an s-shape with the bend corresponding with the Néel transition temperature. This higher temperature rhombohedral ($R\bar{3}c$, No. 167) structure is a $2^{1/2}a_p \times 2^{1/2}a_p \times 12^{1/2}a_p$ unit cell expansion of the primitive cubic perovskite lattice.

A second phase transition occurs at approximately 1867 ± 30 °C from rhombohedral ($R\bar{3}c$) to cubic ($Pm\bar{3}m$) (Figure 3). This transition is second order in nature and very close to the material's melting temperature (1890 °C), and therefore, it has not been experimentally fully investigated. It must be noted that there have been only a few studies regarding LaFeO_3 , while many doped derivatives have been widely studied. Figure 5 shows cations that have been successfully added to LaFeO_3 and which sites (La or Fe) they were determined to occupy.^{7,25–58} To our best of knowledge, the local structure of undoped LaFeO_3 has not yet been investigated, most likely due to the complexity of magnetic neutron total scattering.

Role of A-Site Doping in LaFeO_3 Phase Properties.

While LaFeO_3 is inferior compared to its doped counterparts, it remains critical to understand both the parent structure and how it changes when doped with different cations. The alkaline earth cations (Ca^{2+} , Sr^{2+} , and Ba^{2+}) are the most common A-site dopants. The ionic radius of calcium is very close to that of lanthanum, while strontium and barium are much larger. The most common dopant used to manipulate the ionic conductivity of this material is Sr^{2+} ,^{7,34,35,37,38} which has a larger ionic radius (~ 0.1 Å larger) and forms more ionic bonds with oxygen. LaFeO_3 has two compensation mechanisms that can take place when the material is substituted with lower valence cations, the iron can change its valence, oxygen vacancies can be created, or some combination of both. Therefore, the incorporation of these dopants is typically used to control the ionic conductivity of the material. Strontium has a larger solubility limit and increases the oxide ion conductivity more than that of calcium and barium members.

When Sr is doped into the LaFeO_3 structure ($\text{La}_{1-x}\text{Sr}_x\text{FeO}_{3-\delta}$), it appears to randomly disperse on the A-site of the perovskite lattice, as no cation or oxygen vacancy ordering is observed. The solubility of Sr in LaFeO_3 is quite unique as there are several phase transitions and two-phase regions involved; this is shown in Figure 6. The orthorhombic ($Pnma$) structure manifests between $0 \leq x \leq 0.2$, after which a two-phase region spans until $x = 0.4$ where both the orthorhombic and rhombohedral phases exist.³⁴ The addition of Sr continuously shrinks the unit cell over the orthorhombic region indicating, that Fe has oxidized or oxygen vacancies have been induced. The rhombohedral phase has been shown as phase pure between $0.40 \leq x \leq 0.6$. A second mixed phase region is observed between $0.7 < x < 0.8$ where the rhombohedral and cubic phases both exist. Finally, the structure takes a cubic form between $0.8 \leq x \leq 1.0$ in $\text{La}_{1-x}\text{Sr}_x\text{FeO}_{3-\delta}$.

The primitive cubic edge length can be seen to decrease with increasing Sr content in $\text{La}_{1-x}\text{Sr}_x\text{FeO}_{3-\delta}$ (Figure 6). The

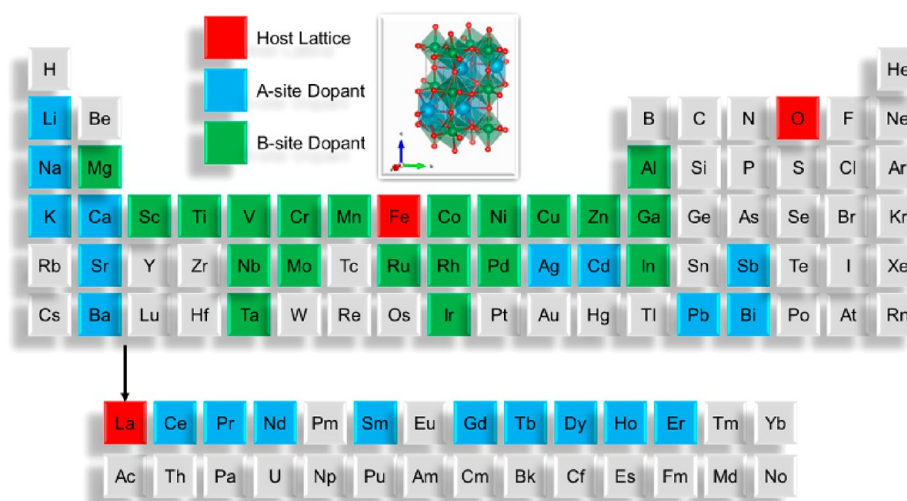


Figure 5. LaFeO₃ lattice that can be doped on both the A- and the B-sites by a wide selection of cations. The [Table 3](#) shows what has been already doped into LaFeO₃.

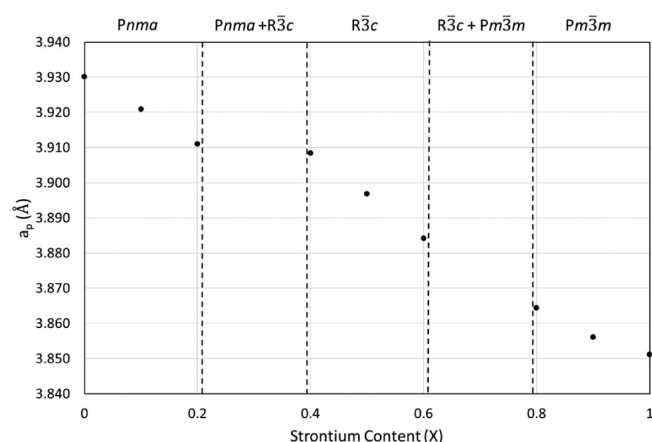


Figure 6. Phase diagram of strontium doped lanthanum ferrite, $\text{La}_{1-x}\text{Sr}_x\text{FeO}_3$, at room temperature. Strontium content is plotted against the primitive cubic unit cell edge length. Three distinct phase pure regions and two mixed phase regions can be observed.

addition of Sr changes the average bond lengths between the A-site and oxygen, which get longer, and the B-site and oxygen, which shortens them. The experimental bond lengths can be

seen in Table 3. Ca-doping in LaFeO_3 offers a unique opportunity to study the effect of defects on the structure without changing the size of the A-site cation.^{54–57} The solubility limit of Ca in LaFeO_3 is controversial, but values up to $x = 0.67$ have been reported. The structure retains the orthorhombic ($Pnma$) phase through the entire solid solution, and the unit cell volume decreases in a linear fashion. The overall dc electrical conductivity is enhanced by several orders of magnitude with increasing Ca content, and at $x = 0.20$ the conductivity reaches a value of $7.8 \text{ S}\cdot\text{m}^{-1}$ at 800°C .⁵⁴ The bond length between the A-site and oxygen increase with Ca doping until $x = 0.2$ at which point this bond starts to decrease, while the B-site to oxygen bond length decreases with increasing Ca content in $\text{La}_{1-x}\text{Ca}_x\text{FeO}_{3-\delta}$ (Table 3)

The Ba-substituted $\text{Ba}_{0.3}\text{La}_{0.5}\text{FeO}_3$ crystallizes in the primitive cubic space group $Pm\bar{3}m$ (No. 221) with a unit cell edge length of $a = 3.935 \text{ \AA}$ and one formula unit per unit cell ($Z = 1$).³⁸ The A-site cations have face-sharing polyhedral units. Neither the A-site or B-site cation ordering has been observed. The addition of Ba to the LaFeO_3 lattice greatly increases the A-site to oxygen bond length in order to accommodate the larger cation, and the average B-site to oxygen bond length does shorten a small amount (Table 3).

Table 3. Average Bond Lengths between Iron and Oxygen that Can Be Observed to Shrink as Any Group II Cation Is Doped on the Lanthanum Site while the Average Bond Length between the A-Site (La and Ca, Sr, or Ba) and Oxygen Increases as a Function of Cation Size and Cation Mismatch (Ca First Grows and then Shrinks after $x = 0.20$)

composition	average Fe—O bond length, Å	average A—O bond length, Å	Fe—O—Fe bond angles, deg	standardized volume, Å ³	ref
LaFeO ₃	2.006	2.605	156.3, 157.3	60.756	21
La _{0.90} Ca _{0.10} FeO ₃	1.995	2.604	155.9, 159.2	60.115	55
La _{0.80} Ca _{0.20} FeO ₃	1.989	2.610	157.2, 160.0	59.821	55
La _{0.667} Ca _{0.333} FeO ₃	1.983	2.593	157.4, 158.5	58.817	56
La _{0.333} Ca _{0.667} FeO ₃	1.946	2.557	157.9, 160.3	56.205	57
La _{0.9} Sr _{0.10} FeO ₃	1.983	2.623	166.7, 162.71	60.3975	34
La _{0.80} Sr _{0.20} FeO ₃	1.976	2.646	172.9, 163.4	59.9900	34
La _{0.60} Sr _{0.40} FeO ₃	1.962	2.681	166.32	59.1817	34
La _{0.50} Sr _{0.50} FeO ₃	1.953	2.688	168.70	58.8117	34
La _{0.40} Sr _{0.60} FeO ₃	1.945	2.745	171.35	58.4183	34
La _{0.30} Sr _{0.70} FeO ₃	1.937	2.738	175.69	58.0267	34
La _{0.5} Ba _{0.5} FeO ₃	1.968	2.783	180	60.948	38

The A-site has also been doped/substituted with Li, Na, K, Ag, Cd, Ce, Pr, Nd, Sm, Gd, Dy, Pb, and Bi.^{25,36,43,44,50–52,59} This demonstrates that LaFeO₃-based materials exist in a highly versatile compositional space which allows a large degree of structural modifications.

Role of B-Site Doping in LaFeO₃ Phase Properties.

The B-site of LaFeO₃ is commonly doped with transition metal cations due to similar ionic radii between the dopant and iron. The effects of Cr, Mn, Co, Ni, and Cu have been extensively studied; as well, the effects of Mg, Al, Sc, Ti, V, Ga, Nb, Mo, Ru, Rh, Pd, In, Ta, and Ir have also been investigated.^{26,40–42,47,53,58,60–65} The most common dopants are Co and Ni as these late transition metals can change the structure from orthorhombic to rhombohedral. The solid solution between LaFeO₃ and LaNiO₃ is smooth with a composition phase transition around the 50–50 composition from *Pnma* to *R3c*.^{30,31} This composition commonly displays both phases due to a first order phase transition near room temperature; the structure is orthorhombic below ~27 °C and rhombohedral above ~27 °C.³¹ The most studied composition is LaFe_{0.8}Co_{0.2}O_{3–δ} due to its relationship to the codoped high-performance phase La_{0.6}Sr_{0.4}Co_{0.2}Fe_{0.8}O_{3–δ}.

When LaFeO₃ is doped with Co, the structure remains orthorhombic up to $x = 0.40$ in LaFe_{1–x}Co_xO_{3–δ} and transitions from orthorhombic to rhombohedral at about the 50–50 composition. A small multiphase region spans between $0.40 < x < 0.60$ where both the rhombohedral and orthorhombic phases coexist.^{28,29} LaFeO₃ has been codoped with both Sr and Co with the following chemical formula, La_{0.6}Sr_{0.4}Co_yFe_{1–y}O_{3–δ} ($y = 0.1, 0.2, \text{ and } 0.3$), in order to enhance both the ionic and the electronic conductivities.^{66–70} The most studied composition is La_{0.6}Sr_{0.4}Co_{0.2}Fe_{0.8}O_{3–δ}, which has a decent p-type electronic conductivity and high catalytic activity for oxygen exchange.^{2–4} This material has also been studied at temperatures as high as 1000 °C in SOECs for CO₂ reduction.² The thermal stability of this composition was investigated by coulometric titrations as a function of partial oxygen pressure to elucidate the redox state. The results indicate that this composition is stable at 750 °C and 10^{–5} atm.³ The structure remains rhombohedral at 1000 °C, which may be an indication that the rhombohedral structure has a relationship with the observed properties.

The effects of Cr and Mn should also be mentioned as they are beneficial to dope multivalent transition metals into the structure to assist with the catalytic function of CO₂ reduction. A full solid solution between LaFeO₃ and LaCrO₃ can be achieved.⁷¹ These compositions crystallize in the orthorhombic space group *Pbnm*. The main composition of interest is LaFe_{0.5}Cr_{0.5}O₃ as the conductivity reaches a maximum.⁷² This composition also displays excellent oxygen evolution reaction (OER) performance.⁷³ The addition of manganese, LaFe_{1–x}Mn_xO₃, drives a disorder induced localization of charge carriers. This in turn increases the overall electrical conductivity as a function of Mn content, up to $x = 0.5$.⁷⁴ The catalytic performance of Mn-substituted LaFe_{1–x}Mn_xO₃ samples exhibits a low charge transfer resistance and abundant oxygen ion species on the surface of the material to enhance its catalytic performance.⁷⁵

To summarize, the A-site is typically doped with alkaline earth ions to control the ionic conductivity of the doped LaFeO₃ material. A necessary modification is to make sure the reduction of CO₂ is driven in an efficient manner, as the diffusion of oxygen through the crystal structure is limited.

Doping Sr is typically preferred over Ca and Ba due to higher overall conductivities, which is intensified by higher solubility limits and disorder caused by cation ionic radii mismatch. The B-site is responsible for the catalytic sequestration of CO₂ and electron transfer that initiates the reduction of CO₂ into useful chemical streams. The usage of multivalent cations such as Cr or Mn on the B-site helps to improve both electronic conductivity of the material and electron transfer between the electrode material and the feedstock gas.

3. OXIDE ION TRANSPORT IN LaFeO₃-BASED PEROVSKITES

A high oxygen conductivity is very important for efficient SOEC operation; thus, the enhancement of the oxide ion conductivity and a better understanding of the transport mechanisms within the crystal lattice are critical. This is especially important for MIEC catalysts, which often have much lower ionic conductivity, compared to solid electrolytes. Recent studies have been mainly focused on improving the transport properties, especially for perovskite-type structure compounds.⁷⁶ Oxygen migration in a solid is typically driven by oxygen vacancies, which are present within the crystal lattice. There are two important parameters that can explain the oxide conductivity in oxides, which include the oxygen vacancy concentration and the vacancy migration energy barrier. Both are experimentally measurable and commonly used in computational methods to predict the impact of dopants on the material's conductivity.⁷⁷ For a more detailed discussion regarding the success of computational modeling involving molecular dynamics and DFT of perovskite-type MIECs, the reader is referred to the papers published elsewhere.^{78,79}

DFT+U is often used to predict the formation energy of oxygen vacancies. Lee et al. were able to calculate the formation energy values and defect concentration for LaMnO₃ in temperatures ranging between 600 and 1200 °C.⁸⁰ The obtained data was in good agreement with the experimental values reported previously. In addition, valuable information regarding the defect chemistry and oxygen tracer diffusivity was obtained. Another study demonstrated that single crystals of LaFeO₃ and LaCoO₃ had a higher tracer diffusion coefficient (D_0) for oxide ions when the concentration of the oxygen vacancies was increased. The proportional relationship between D_0 and the oxygen partial pressure (p_{O_2}) was established. These findings lead to the conclusion that the oxygen diffusion in this perovskite structure follows a vacancy migration mechanism. It was also shown that increasing the substitution of La³⁺ with Sr²⁺ led to either more Fe⁴⁺/Co⁴⁺ or more oxide ion vacancies, and an increase in the oxide ion conductivity was observed. Sr-doping in LaFeO₃ is a well-established method for increasing the oxygen ion conductivity.^{81,82} In both LaFeO₃ and LaCoO₃, vacancy formation is observed with a dominating redox process and a lower vacancy formation energy.^{83–85}

The oxide ion vacancy formation energy, migration energy, and diffusion (D) can be related by expression 2.⁸⁴

$$D = A \cdot \exp\left(-\frac{\Delta H_{f,vac} + \Delta H_{mig}}{k_b T}\right) \quad (2)$$

where A is the pre-exponential term, $\Delta H_{f,vac}$ is the enthalpy of oxygen vacancy formation, ΔH_{mig} is the enthalpy of the oxide ion migration, k_b is Boltzmann's constant, and T is the absolute

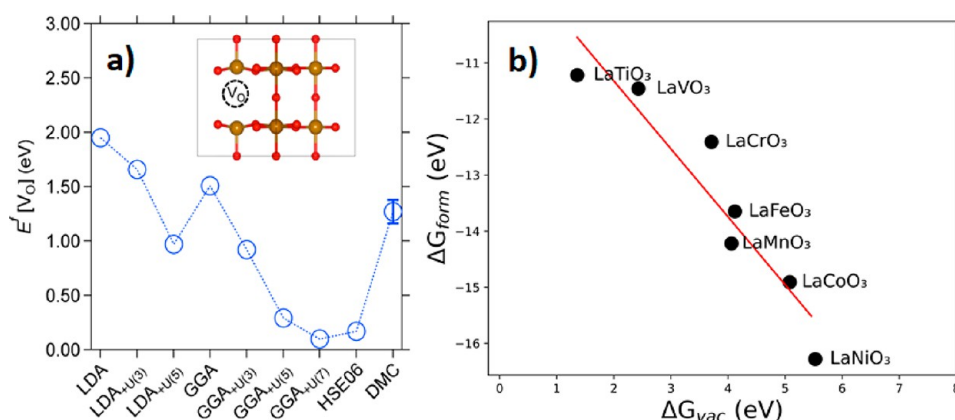


Figure 7. (a) Calculations of vacancy formation for LaFeO₃ and (b) the linear relationship between ΔG_{Form} and ΔG_{Vac} for the structure of LaMO₃ (M = transition metals) perovskites. Adapted with permission from ref 86. Copyright 2017 AIP Publishing.

temperature. Computational methods to calculate the enthalpies include the classical potential approach from Jones and Islam⁸³ and quantum mechanical methods such as DFT+U^{77,85} or quantum Monte Carlo.⁸⁶ The classical potential approach is a very efficient method as the calculation is reduced to the long-range Coulombic repulsion and the short-range Born potential treating repulsion and van der Waals attraction,⁸⁷ $\varphi_{ij}(r) = A_{ij} \exp(-r/\rho_{ij}) - (C_{ij}/r^6)$, and the results were calculated using the GULP code,⁸⁸ which can also be used to calculate polymers.⁸⁹ Quantum mechanical methods such as DFT+U treat valence electrons individually, and hence, this technique requires more time. These techniques are used to calculate energy barriers between two energy minima to obtain the diffusion constant. Dealing with defect structures makes it more expensive to run with DFT.⁸⁶ Pure DFT calculations do not include kinetic terms which are important for a diffusion pathway investigation.⁸⁷ To investigate the diffusion, a combination of classical potentials and DFT calculations such as the diffusion Monte Carlo (DMC) method has been applied.⁹⁰ The atoms of interest are treated with a many electron model while everything else is simulated using classical potentials.

Considering there are two distinct crystallographic positions for oxygen, each oxygen position has the potential to migrate under a different mechanism in LaFeO₃. The first oxygen position (O1) lies on the same level as the lanthanum cations, and oxygen in this position acts as an axial oxygens in the iron octahedra; the other oxygen position (O2) acts as the equatorial oxygen positions in the layers of iron cations. The vacancy formation energy only differs by 0.04 eV between two oxygen positions.⁷⁶ Thus, oxygen diffusion can follow three different pathways (from O1 to O1, O2 to O2, or O1 to O2). Calculations indicate that the diffusion from O1 to O2 gives the lowest enthalpy. The calculated bond distances and lattice parameters that allowed the orthorhombic structure to be reproduced were found to be in a good agreement with experimental values. In another study, DFT+U showed similar pathways for the O1–O1 and O2–O2 and a different pathway was obtained for the O1–O2.⁷⁷

The process of calculating the migration barriers and vacancy formation energies for all possible perovskite-type structure compounds is very cumbersome (Figure 7)⁸⁶ and time-consuming. The high magnetization of Fe ions and antiferromagnetic structure of LaFeO₃ make it challenging for DFT calculations. Recent studies include Hubbard U

corrections which utilize optimized U parameters against experimentally measured lattice parameters and magnetization.^{84,85,91} Comparing vacancy formation energies or migration barriers in one compound using different computational methods can lead to great discrepancies, and limited information is inferred. However, using the computational methods, trends can be derived and could be related to the experimental study.

The stability of lanthanum-based perovskites with different doping ratios shows a relationship between vacancy formation energy and bulk formation energy of the compound.⁹² Su and Sun investigated the formation energy of oxygen vacancies and found a linear relationship between the bulk formation energy and the formation energy of oxygen vacancies in ABO₃ (A = La and/or Be, Mg, Ca, Sr, and Ba; B = Ti, V, Cr, Mn, Fe, Co, and Ni).⁹⁴ This indicates that the vacancy formation energy rises inversely proportionally linear with the formation energy of the bulk. To confirm the linear relationship, they repeated their calculations on LaFe_{0.5}Mn_{0.5}O₃, and they found a similar slope and intercept values.

An extensive study on LaMO₃ (M = Sc–Cu) perovskites using a different DFT indicates a relationship with the migration of oxygen along the periodic table.⁷⁷ This study shows that LaFeO₃ is an exception within this series with a higher vacancy formation energy than expected from experimental studies (Figure 8). This discrepancy originates from having a significantly lower exchange stabilization energy upon reduction of Fe, as the additional electron takes an antiparallel spin as all five d-orbitals are occupied by a single electron. DFT+U calculated vacancy formation energies are in

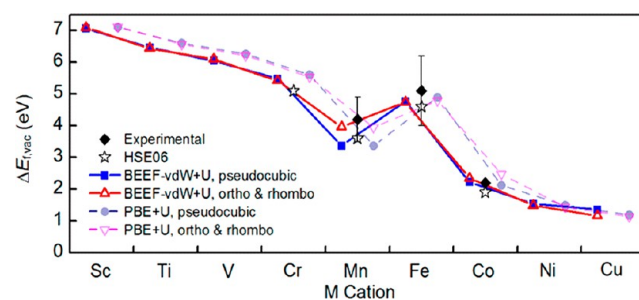


Figure 8. Calculated $\Delta E_{f,vac}$ in LaMO₃ by using the BEEF-vdW+U and PBE+U methods. Reproduced with permission from ref 77. Copyright 2019 American Chemical Society.

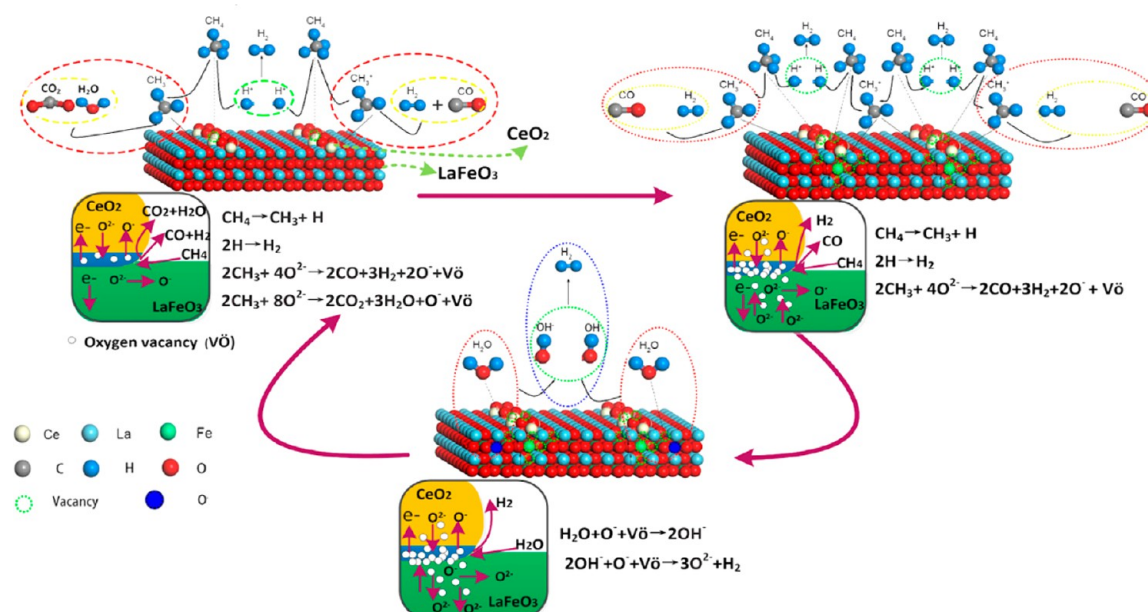


Figure 9. Reaction mechanism model for chemical-looping reforming of methane (CLRM) over $\text{CeO}_2/\text{LaFeO}_3$ oxygen carriers. Reprinted with permission from ref 93. Copyright 2017 Elsevier.

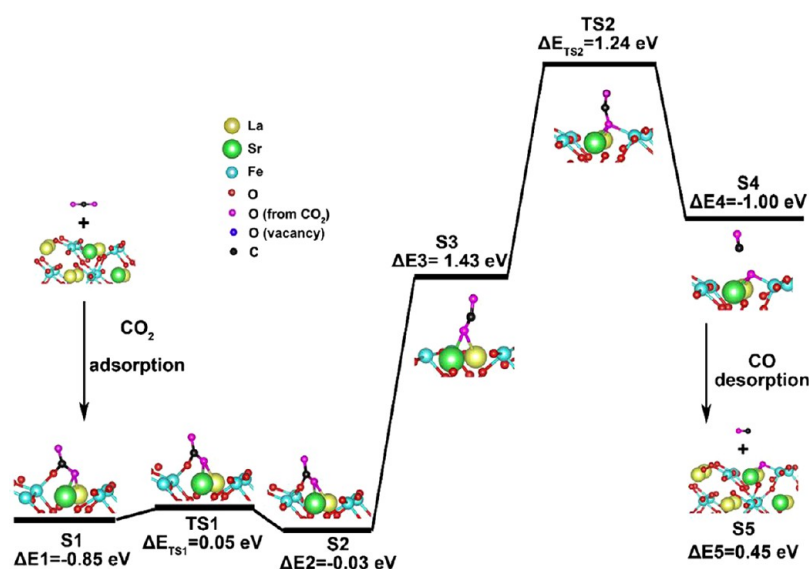


Figure 10. Proposed pathway for CO_2 reduction through carbonate species. Reprinted with permission from ref 7. Copyright 2018 Elsevier.

good agreement with the experimental results. The observed trend of the oxygen migration barrier can be described by the oxygen vacancy formation energy, as a descriptor. The effect of macropores in LaFeO_3 as an oxygen carrier for chemical-looping reforming of methane was examined by Zheng et al.⁹³ This process allows the production of H_2 and CO with a $\text{H}_2:\text{CO}$ ratio of 2. The authors suggest that the coexistence of $\text{Ce}^{4+}-\text{Ce}^{3+}$ and $\text{Fe}^{3+}-\text{Fe}^{4+}$ couples on the interface of LaFeO_3 and CeO_2 provides many oxygen vacancies as active sites for methane adsorption and activation (Figure 9).

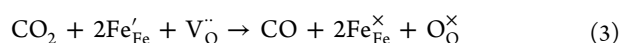
4. ELECTROCHEMICAL PERFORMANCE OF LaFeO_3 PEROVSKITES

Electrochemical CO_2 Reduction on LaFeO_3 -Based Materials. The electrochemical reduction of CO_2 can proceed via multielectron reaction pathways and yield many different

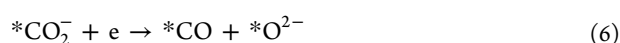
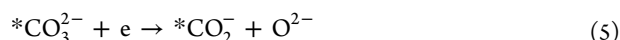
reaction products including carbon monoxide, formic acid, formaldehyde, methanol, and methane.^{94,95} While these products can be obtained using low temperature electrolysis systems, they are limited by low CO_2 mass transfer to the surface of the cathode, and thus, low current densities are obtained.^{96,97} Hence, research efforts have been aimed toward high temperature SOFCs to overcome this issue. A higher operating temperature is beneficial for increasing the kinetics and/or entropy change of the reaction.⁹⁸ Especially for water, the product significantly increases its entropic contribution from 25 to 100 °C, together with the CO contribution, its product outweighs the entropy of CO_2 and H_2 , leading to a more thermodynamically favorable situation for the formation of CO . The ΔG formation energy for $\text{CO} + \text{H}_2\text{O}(\text{g})$ from $\text{CO}_2 + \text{H}_2$ equals 0 at 706 °C and becomes negative at higher temperatures. Unlike LaCoO_3 and LaNiO_3 which are reduced

by H_2 , LaFeO_3 and LaMnO_3 are more stable under the reducing condition at temperatures greater than 300 °C.⁹⁹ EXAFS/XANES studies confirmed the trivalent oxidation state of B-site cations after treatment under H_2 , while the CO yield was reported as 22.8% for LaFeO_3 at 320 °C and 8.3% for LaMnO_3 at 380 °C.

An electrochemical study on $\text{La}_{0.8}\text{Sr}_{0.2}\text{CuO}_4$ showed the formation of methanol, ethanol, and propanol with a faradic efficiency of up to 40% with a current density of 180 A/cm².¹⁰⁰ In optimizing the catalytic activity, two main strategies have been carried out: (i) elemental doping and (ii) structural modification. Hu et al. studied $\text{La}_{0.8}\text{A}_{0.2}\text{FeO}_{3-\delta}$ (A = Ca, Sr, and Ba) perovskites as cathodes for CO_2 reduction.⁸ Doping with alkaline earth elements decreases the average valence states of Fe, facilitating electron transfer in the reduction of CO_2 via the surface reaction 3. A 20% Sr-doping was found to be the optimal composition for the promotion of CO_2 reduction.



In order to further advance the development of novel catalysts for the CO_2 reduction reaction (CO_2RR), a detailed understanding of the reaction mechanisms at the cathode $\text{LaFeO}_{3-\delta}$ is crucial. Yang et al. investigated the mechanism of CO_2RR on LSF through Raman spectroscopy.⁷ Their findings were corroborated by density functional theory (DFT) calculations, which revealed the lowest energy pathway for the CO_2RR was through a carbonate species (Figure 10).⁷ This carbonate species then goes through a decomposition to form CO (rate-determining step) followed by the subsequent desorption of CO. The proposed mechanism can be written as follows:⁷



Opitz et al. investigated the surface reactions occurring on $\text{La}_{0.6}\text{Sr}_{0.4}\text{FeO}_{3-\delta}$ and $(\text{La},\text{Sr})(\text{Cr},\text{Ni})\text{O}_{3-\delta}$ thin films by means of in operando near-ambient pressure X-ray photoelectron spectroscopy (NAP-XPS).¹⁰¹ Under cathodic polarization conditions, CO_2 is activated through the formation of a carbonate species. The carbonate species was found to form through the adsorption of CO_2 on an oxygen vacant site followed by an electron transfer step (Figure 10). These mechanisms suggest that perovskites with both high electron concentrations and oxygen vacancy concentrations are well suited for CO_2RR . The presence of oxygen vacancies in perovskite-type materials has been shown to not only favor this ion conduction but provide a site of adsorption for intermediates such as CO_2 .

Main Approaches for Electrochemical Performance Enhancement. From a thermodynamic point of view, it is more profitable to run SOECs at a high temperature due to a lower electrical energy demand, decreased reaction over potential, and faster oxygen ion diffusion.^{102,103} In practice, high operation temperatures not only limit the range of materials that can be used for cell production but also increase the costs of cell manufacturing and operation. High temperature operation dramatically enhances the material degradation

rate.¹⁰⁴ The obvious way to increase the long-term stability and decrease cell manufacturing costs is to reduce the cell operation temperature. The usage of a composite electrode helps not only to overcome the high activation energy issue but also to reduce incompatibility between components due to the mismatch between thermal expansion coefficients as well as decrease the particle coarsening rate. This approach also helps to combine different materials to eliminate their disadvantages and thus synergistically increase preferred operational characteristics. There are two different ways to make the composite electrodes that are widely used by researchers in SOCs.

Mechanical Mixing. The first approach is mechanical mixing of the individual components, predominantly the catalyst and the electrolyte, followed by surface deposition of the obtained slurry. Xi et al. have shown that the highly active composite $\text{La}_{0.6}\text{Sr}_{0.4}\text{Co}_{0.2}\text{Fe}_{0.8}\text{O}_{3-\delta}$ (LSCF)– $\text{Ce}_{0.9}\text{Gd}_{0.1}\text{O}_{2-\delta}$ (GDC) electrode can be obtained by mechanical processing of La_2O_3 , $\text{Sr}(\text{OH})_2$, Co_3O_4 , Fe_2O_3 , and GDC powders within a short period of time (around 20 min) and then sintering at 900 °C for 2 h.¹⁰⁵ The symmetrical half-cell tests utilizing a dense GDC layer as an electrolyte in air have shown very low polarization resistances of 0.08, 0.17, 0.38, and 0.69 Ohm·cm² at temperatures of 800, 750, 700, and 650 °C correspondingly. Those values are close to the previously reported high-performance composite cathodes, obtained by the more expensive and time-consuming precursor infiltration method.¹⁰⁶

Another highly active electrode, made by the mechanical mixing approach, is the LSCF–BSCF ($\text{Ba}_{0.5}\text{Sr}_{0.5}\text{Co}_{0.8}\text{Fe}_{0.2}\text{O}_{3-\delta}$) composite. This composite is a typical example of how two materials can be combined to eliminate disadvantages, by utilizing their individual strengths. LSCF has high electrical conductivity but a low oxygen diffusion rate.^{107,108} In contrast, BSCF has a high oxygen diffusion rate and catalytic activity¹⁰⁹ but low chemical stability.¹¹⁰ Giuliano et al. combined these materials in a 1:1 volume ratio.¹¹¹ The XRD analysis showed that, after sintering the starting perovskites, Co-rich $\text{La}_{0.5}\text{Ba}_{0.5}\text{CoO}_3$ and Sr-rich $\text{La}_{0.4}\text{Sr}_{0.6}\text{FeO}_3$ phases were formed. This electrode was then tested in the symmetrical $\text{Ce}_{0.8}\text{Sm}_{0.2}\text{O}_2$ (SDC) supported cell configuration along with LSCF and BSCF as reference cells. It was shown that the composite materials cell had a higher polarization resistance than the pure-BSCF cell, but it was lower than that of the LSCF electrode (0.07, 0.036, and 0.21 Ohm·cm² at 700 °C, respectively). Nevertheless, the degradation rate for the LSCF–BSCF cell is significantly lower—only 5% over 200 h—whereas it was 29% for pure LSCF over 72 h and 38% for pure BSCF over 200 h. A lower activation energy was achieved for the composite cell and by comparison is much lower than that of the pure BSCF electrode (96 vs 113 kJ·mol⁻¹).

Ghosh et al.¹¹² produced functionally graded composite interlayer structures based on an LSCF–LSFC ($\text{La}_{0.6}\text{Sr}_{0.4}\text{Fe}_{0.2}\text{Co}_{0.8}\text{O}_3$) 1:1 mixture (CF-1) and Co and Gd-codoped ceria (CoGDC). Varying the ratio of CF-1 (L) and CoGDC (C) yielded layers with different properties. Figure 11 shows the XRD patterns for each of the compositions made as well as for the pure components. At 800 °C, the lowest electrode polarization resistance of 0.04 Ohm·cm² was obtained for a mixture of 80:20 L:C. In conjunction with the amount of Co-GDC increasing, the layer polarization resistance also increases up to 166 Ohm·cm² for the 20:80 mixture. On the contrary, by adding more of the L component,

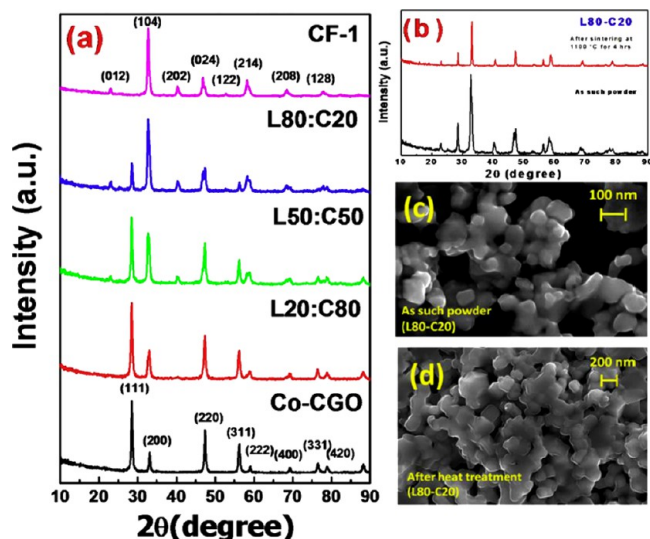


Figure 11. (a) X-ray diffraction patterns of the base materials CoCGO and CF-1 and the composite materials L80-C20, L50-C50, L20-C80, (b) comparison plots of XRD obtained from as such powder and after sintering of the composite L80-C20, and FESEM images of the composite L80-C20 as (c) powder and (d) sintered (at 1100 °C) bulk. Reprinted with permission from ref 112. Copyright 2016 Elsevier.

the thermal expansion coefficient match between the catalyst and the electrolyte decreases, which may lead to quick cell degradation. To avoid this issue, a multilayered structure was proposed, which can be written as L80-C20/L50-C50/L20-C80/CoCGO/L20-C80/L50-C50/L80-C20. For such a case the resistance was as low as 0.067 Ohm·cm². The cell degradation for that condition was found to be about 2.9% after 600 h of operation.

Infiltration Method. The second approach of producing a composite electrode is the precursor infiltration method. In this case, cation containing salts, predominantly nitrates, are mixed together with surfactants, chelating, and complexing agents to form a polymeric precursor solution.^{113,114} This solution is then infiltrated inside the porous structures and sintered after to form the desired phase and structure (Figure 12). Even though this approach is much more costly and time-consuming, it allows control of the catalyst particle size as well as backbone coverage, so materials with precisely controlled structures and extremely high activity can be produced. The influence of surfactants and chelating agents on the composite structure properties was shown by Burye and Nicholas.¹¹⁵ It was demonstrated that the preinfiltration of GDC nanoparticles decreases the average size of LSCF, which is subsequently infiltrated by a nitrate solution containing a chelating agent (citric acid) or a surfactant (Triton X-100). On the contrary, the average particle size is not observed to decrease when LSCF was infiltrated from a nitrate solution without any additives (Figure 13). Symmetrical cells with Triton X-100-derived LSCF–GDC cathodes in air have shown a polarization resistance of 0.17 Ohm·cm² at 540 °C, while a value of 0.37 Ohm·cm² was obtained for the cathode prepared without any additives. In addition, a much slower degradation rate was observed for systems that utilize either a chelating agent or a surfactant.

Nanoparticles Ex-solution as a New Promising Approach of Catalytic Activity Enhancement. The development of materials with nanoparticles distributed on the carrier surfaces is a well-known approach to produce highly active catalysts in many fields of application. In most of the cases, such nanoparticles are introduced through common deposition methods, such as physical deposition or infiltration.¹¹⁶ Even though these approaches are widely used, they

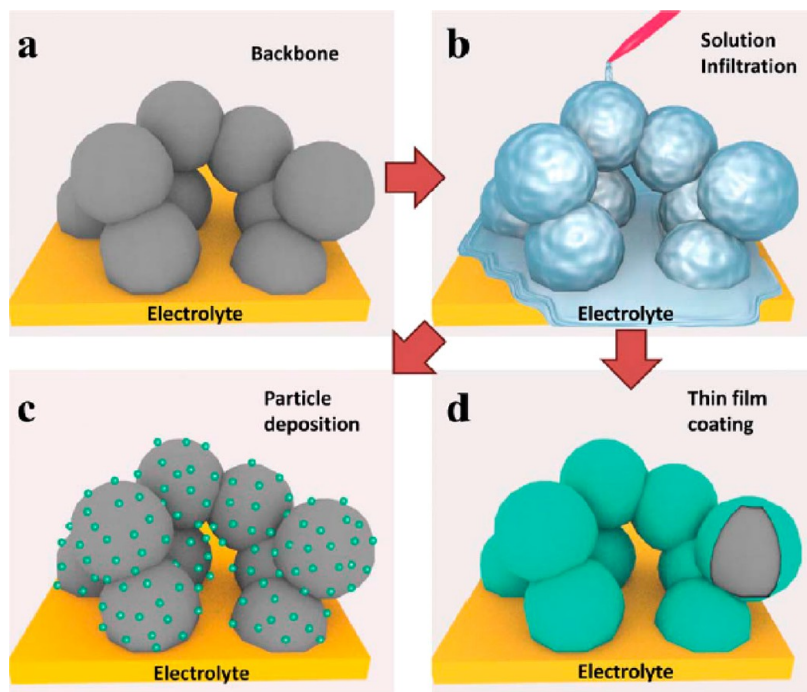


Figure 12. A schematic of a typical infiltration process: (a) an as-fired electrode backbone; (b) a process of solution drops entering into the electrode backbone; and two typical morphologies of infiltrated electrode after thermal treatment: (c) particle deposition and (d) thin film coating. Republished with permission from ref 104. Copyright 2014 Royal Society of Chemistry.

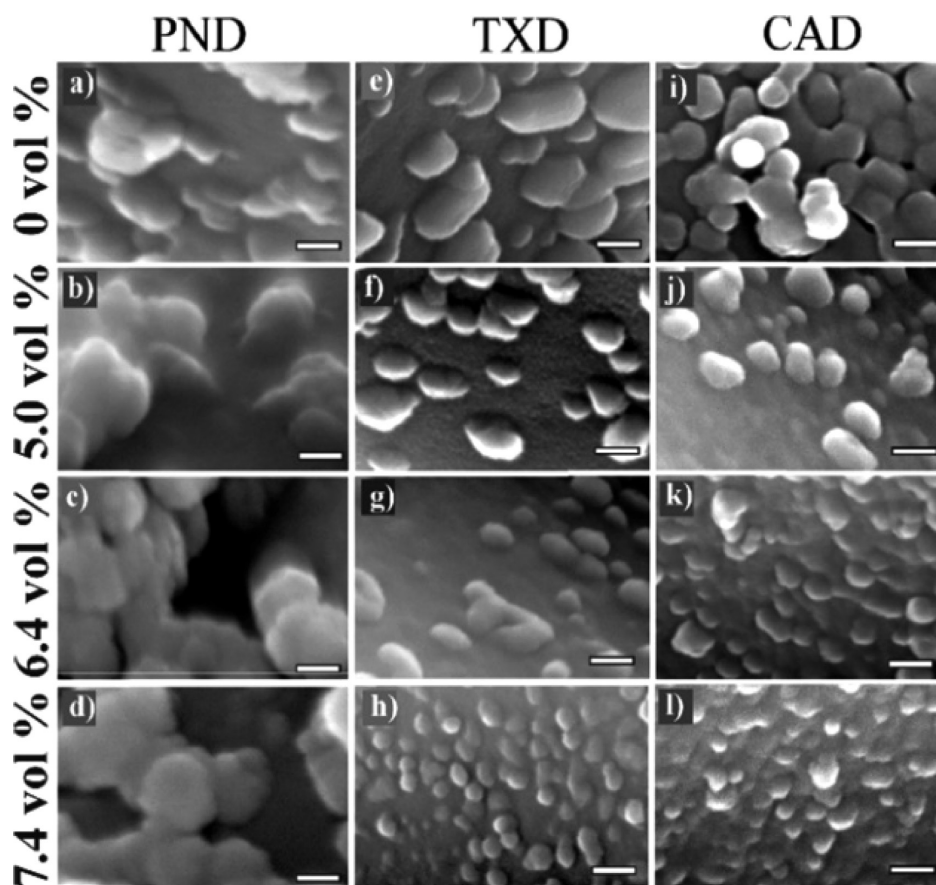


Figure 13. Scanning electron micrographs of 700 °C fired (a–d) 1.50 M LSCF, (e–h) 1.50 M Triton X-100 + LSCF, and (i–l) 0.50 M citric acid + LSCF-GDC cathodes with (a, e, i) no GDC preinfiltration, (b, f, j) 5.0 vol % of GDC preinfiltration, (c, g, k) 6.4 vol % of GDC preinfiltration, and (d, h, l) 7.4 vol % of GDC preinfiltration. Scale bars indicate a length of 50 nm. Reprinted with permission from ref 115. Copyright 2015 Elsevier.

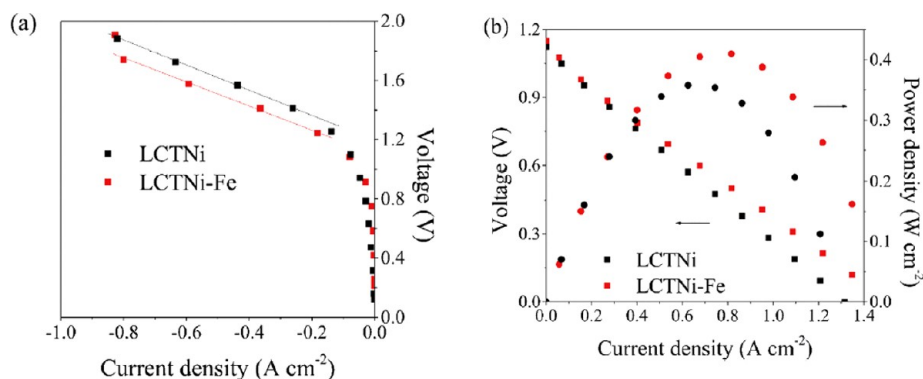


Figure 14. Current density vs voltage and power density curves of a single cell with LCTNi or LCTNi-Fe electrode operated at 900 °C in (a) electrolysis mode in 100% CO₂ and (b) fuel-cell mode in dry H₂. Reprinted with permission from ref 120. Copyright 2019 Elsevier.

only provide limited control over the size of the nanoparticles, distribution, and anchorage of the synthesized species.¹¹⁷ In addition, nanoparticles obtained in such a way tend to segregate, which leads to drop in catalyst activity.¹¹⁸

An ex-solution method to deposit nanoparticles on the materials surface is to grow them directly on the surface in the reducing atmosphere at elevated temperatures. This technique was exhaustively explored by Neagu et al.¹¹⁹ It was shown that nanosize particles, including metals, oxides, and their combinations, can be obtained in a controlled way through deviations from ideal ABO₃ backbone stoichiometry. Through initial composition variation, particle distribution and particle

size can be precisely controlled. It was also shown that even the particles in nanoproximity with respect to each other remained stable even after prolonged heat treatments. It suggests that the particles are firmly anchored on the surface, in contrast to particles obtained by a simple deposition. It was also shown that the nanoparticle ex-solution in electrochemical cells can be reached by the application of an electrical potential.¹²⁰ Specifically, metallic nanoparticles of size 30–100 nm were obtained from La_{0.43}Ca_{0.37}Ni_{0.06}Ti_{0.94}O₃ (LCTNi) and La_{0.43}Ca_{0.37}Ni_{0.03}Fe_{0.03}Ti_{0.94}O₃ (LCTNi-Fe) perovskite phases in a carbon dioxide environment at the reduction potential over 2 V. As can be seen in Figure 14, such cells have shown

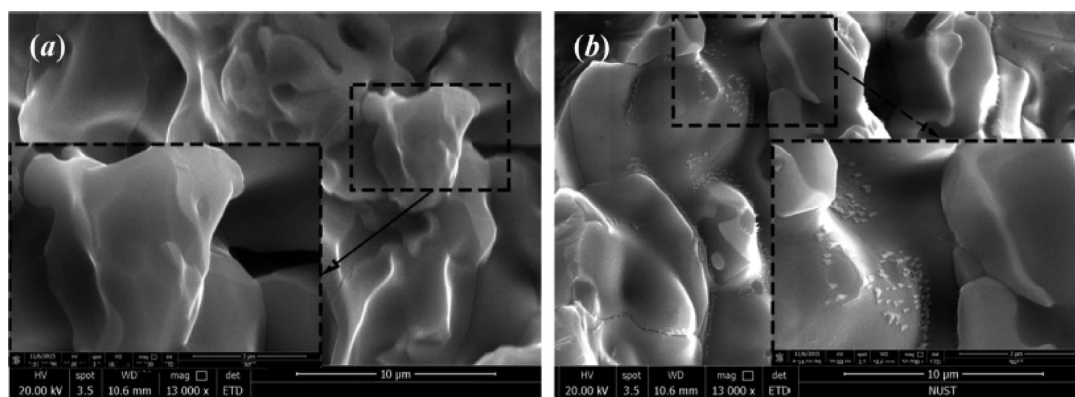


Figure 15. SEM images of (a) pure LCFN ceramic after calcination in air at 1200 °C and (b) LCFN ceramic after further thermal treatment in 10 mL/min H_2 at 800 °C for 10 h. Reprinted with permission from ref 122. Copyright 2017 Elsevier.

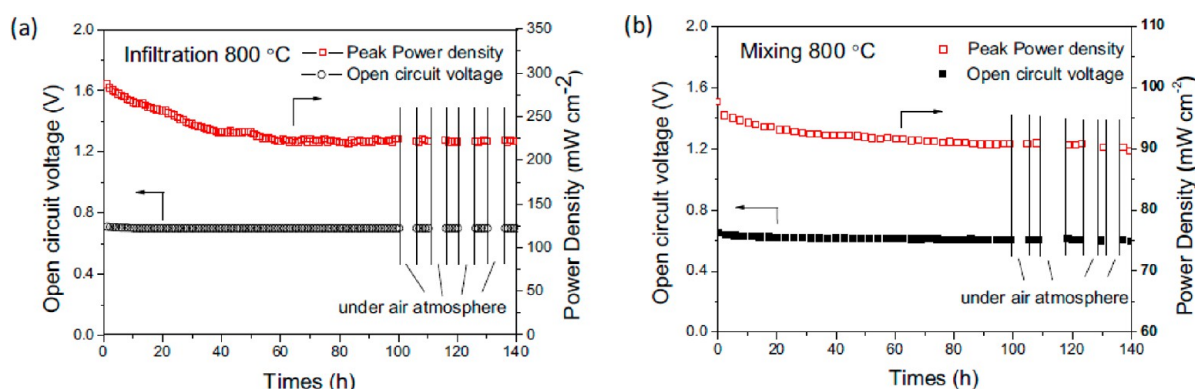


Figure 16. Time dependence of open circuit voltage and powder density of (a) the infiltration and (b) the mixing of the LCFN-30SDC/SDC/ LCFN-30SDC symmetrical cell with dry H_2 as fuel and ambient air as oxidant. Reprinted with permission from ref 122. Copyright 2017 Elsevier.

high electrochemical activity for both H_2 evolution and CO_2RR .

Liu et al. have shown that Fe–Ni nanoparticles were produced from $\text{La}_{0.6}\text{Sr}_{0.4}\text{Fe}_{0.8}\text{Ni}_{0.2}\text{O}_{3-\delta}$ (LSFN) as a perovskite backbone in a reducing atmosphere at 850 °C.¹²¹ The XRD results have shown that the chemical composition of the obtained nanoparticles is $\text{Fe}_{0.64}\text{Ni}_{0.36}$. In addition, some La_2O_3 was formed, which is undesirable due to its low activity. Nevertheless, high electrochemical activity for CO_2RR was obtained as low as 0.272 $\Omega\cdot\text{cm}^2$ at OCV in a 70:30 CO_2 and CO mixture at 850 °C. The stability test over 100 h in the same conditions at 0.6 V (vs OCV) did not show any significant degradation or coke presence. Ding et al.¹²² tested $\text{La}_{0.6}\text{Ca}_{0.4}\text{Fe}_{0.8}\text{Ni}_{0.2}\text{O}_3$ (LCFN)–SDC composites obtained via both infiltration and mechanical mixing with further nanoparticles ex-solution. The presence of both bimetallic Fe–Ni particles with the size of 10–20 nm and GDC nanoparticles with the size around 0.45 μm was confirmed by both X-ray diffraction and scanning electron microscopy (Figure 15). Electrochemical tests in the fuel cell mode have shown that the combination of Fe–Ni and GDC nanoparticles has led to synergistical promotion of both H_2 oxidation and O_2 reduction reactions. Both mechanically mixed and infiltrated electrodes have shown high stability for repeated hydrogen–air cycles (Figure 16). The highest initial power density of 303 $\text{mW}\cdot\text{cm}^{-2}$ was achieved for a cell with infiltrated electrodes while only 140 $\text{mW}\cdot\text{cm}^{-2}$ was achieved for a mechanically produced cell, which dropped to 225 and 90 $\text{mW}\cdot\text{cm}^{-2}$, respectively.

Durability during Electrochemical Operation and Degradation Mechanisms.

In order to increase the ionic conductivity of LaFeO_3 derivatives, the number of oxygen vacancies must be increased. The most common approach is to partially substitute cations on the A-site. Lanthanum (3+) is replaced by an ion with a similar size and lower oxidation state, such as strontium (2+). It has been shown by many researchers that strontium tends to segregate on the surface of the perovskite that is believed to be the main reason for LSCF degradation.^{123–126} In order to better understand the potential role of Sr segregation in the LSCF catalyst degradation mechanism, Oh et al.¹²⁷ investigated the elemental enrichment behavior of this material surface. After the sample went through a heat treatment in air for 50–100 h in the temperature range of 600–900 °C, submicrometer-sized particles had formed on the surface of the grains. Further investigation by Auger electron spectroscopy revealed that the precipitate was composed of strontium and oxygen, and EDX did not show any significant changes in the bulk composition. The amount of precipitate formed increased with an increasing temperature and oxygen partial pressure. A two-step mechanism for the formation of Sr-rich particles was proposed. First, oxygen vacancies are formed, and then precipitation of a Sr-enriched phase occurs. The rate of the first step increases with a decrease in oxygen partial pressure. The second step occurs faster in oxidizing conditions. It is well-known that more precipitants are formed when the oxygen partial pressure increases, thus suggesting that the second step seems to be rate determining.

A similar study was conducted by Yu et al. for LSCF heteroepitaxial thin films.¹²⁸ In this research, perovskite thin films with a varied Sr content were produced by pulsed layer deposition on different single-crystal substrates to obtain different levels of strain inside the film. Strontium segregation was estimated by atomic force microscopy (AFM) and total-reflection X-ray fluorescence. The electronic structure of the Sr-enriched phases was described by hard X-ray photoelectron spectroscopy. It has been shown that adding more strontium to the perovskite lattice leads to higher quantities of strontium segregation (Figure 17). It was found that the formation rate of

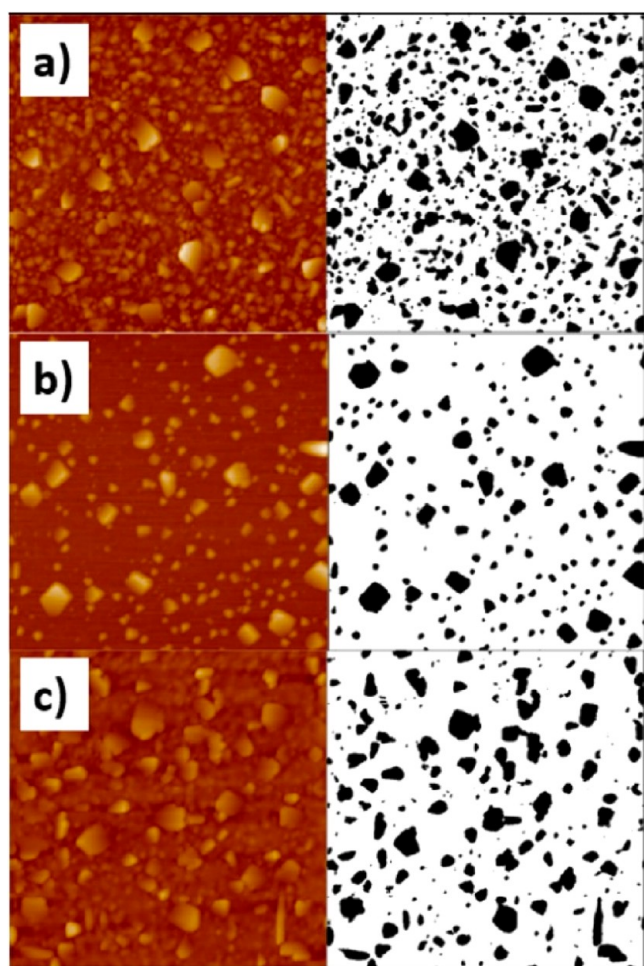


Figure 17. AFM images ($2 \times 2 \mu\text{m}^2$) (left) and corresponding image analysis (right) of surface coverage on postannealed (a) LSCF-6428, (b) LSCF-7328, and (c) LSCF-8228 samples. Reproduced with permission from 128. Copyright 2016 American Chemical Society.

the Sr-rich phase is influenced by lattice strain, which can be introduced by heteroepitaxial growth and can modify the diffusivity of Sr cations within the lattice. The use of high-resolution X-ray diffraction methods revealed that surface segregation is increased with tensile strain and reduced with the compressive strain present in the film. A mathematical model was also developed to correlate the Sr segregation with the relative strain and Sr content within the perovskite phase. The surface segregation increased linearly with the quantity of Sr in the lattice when the relative strain was kept constant. Similarly, the segregation was also increased monotonically

with an increase in relative strain when the quantity of Sr was held constant.¹³⁰

Another factor that can dramatically decrease the activity of Sr-containing perovskite catalysts is Cr poisoning.^{129–131} The most common source of Cr in the cells is the metallic alloys that are used as cell interconnects. Cr might also be introduced by some of the supporting equipment, such as heat exchangers, tubing, and so on. Usually poisoning is caused by the reduction of Cr(VI) species to solid chromium particles, which can deposit on the catalyst surface.¹³² It has been proposed that this reduction is due to a kinetically controlled chemical reaction between gaseous Cr(VI) species and the Sr-rich segregated phases. The most common impurity phases that have been observed include SrCrO_4 and Cr_2O_3 . Large (up to $6 \mu\text{m}$) SrCrO_4 deposits can form and block pores in the catalyst. This is a problem because this phase has a low electrical conductivity and low catalytic activity.

The alternative degradation mechanism has been predicted by Ni et al.¹³³ who observed that higher degradation was achieved for surfaces with homogeneous Cr distribution and low deposited Cr. This is in the contrast to the electrodes which have low Cr contents in the bulk and large amounts of chromium surface deposits.¹³⁴ The degradation due to Cr poisoning of the $(\text{La}_{0.6}\text{Sr}_{0.4})_{0.95}(\text{Co}_{0.2}\text{Fe}_{0.8})\text{O}_{3-\delta}$ (LSCF6428) cathode was investigated in a symmetrical cell mode. EIS data analysis (Figures 18 and 19) have clearly shown that with overall Cr content about 2.7 atom % k_0 and D_0 dropped by 1–2 orders of magnitude. Even at a high poisoning level, focused ion beam scanning electron microscopy (FIB-SEM) tomography did not show any significant changes at the micrometer scale. On the contrary, the incorporation of chromium was clearly seen at the nanoscale level (Figure 20). In addition, the Fe–Co spinel formation was readily increased by Cr poisoning, which might be a consequence of partial B-site cation substitution with chromium. Also, initial evidence of Cr segregation along LSCF grain boundaries was presented, which is associated with a Co deficiency. The methods developed in the presented study provided several new ways to examine the degradation mechanism in SOEC/SOFCs that can be used for testing the stability of new materials operating under real conditions.

Another important issue is that many abundant perovskite structures are not chemically compactable with the conventional yttria stabilized zirconia (YSZ) electrolyte. Sr-containing perovskites, such as LSCF, in contact with YSZ at operation temperatures, form highly resistive $\text{La}_2\text{Zr}_2\text{O}_7$ and SrZrO_3 .⁶² In order to overcome this issue, Uchida et al. have introduced an SDC buffer layer between the YSZ electrolyte and the perovskite catalyst.¹³⁵ In this case, the porosity of the barrier layer determines the cell performance.¹³⁶ In addition, less conductive phases, such as $\text{Ce}_{0.37}\text{Zr}_{0.38}\text{Gd}_{0.18}\text{Y}_{0.07}\text{O}_{1.87}$, are formed due to YSZ and GDC/SDC interdiffusion.¹³⁷ Matsui et al. have described microstructural changes of the LSCF/SDC/YSZ system upon long exposure times with constant current density at 1000°C .¹³⁶ Before the exposure, SEM showed that only the solid solution between SDC and YSZ had formed. After 100 h of exposure time, the structure became more complicated due to SrZrO_3 phase formation, which was confirmed by EDS. With an increase in the time exposed, the amount of SrZrO_3 increased. It was noted that the new phase formation occurs only at the SDC/YSZ interface and only inside the pores, so it was suggested that the formation of SrZrO_3 is attributed to Sr diffusion to the SDC/YSZ interface.

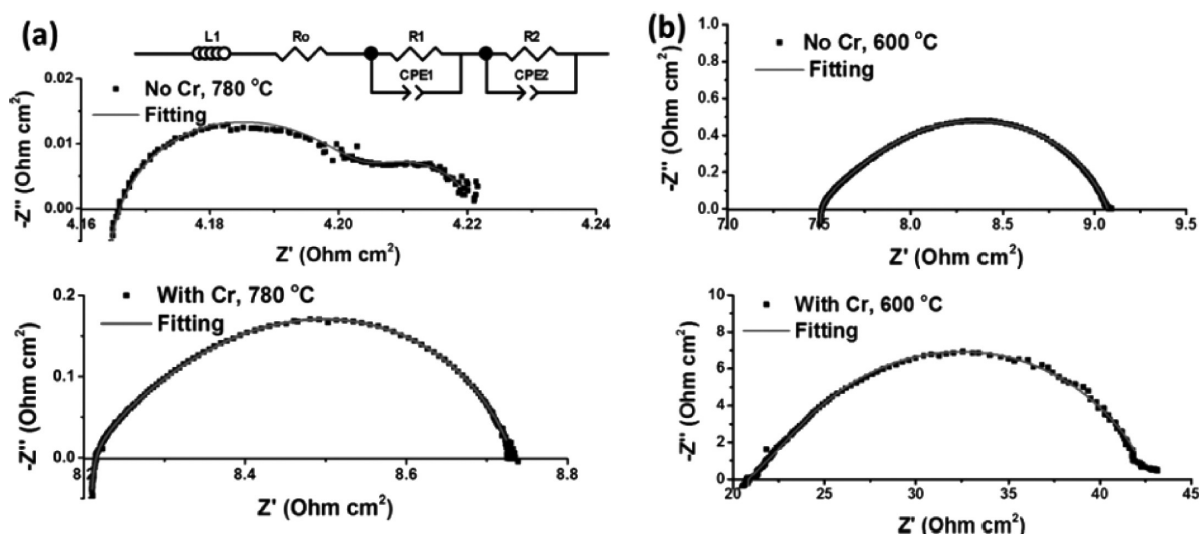


Figure 18. Representative impedance spectra of LSCF6428 cathodes with/without Cr poisoning at (a) 780 °C and (b) 600 °C. The impedance is normalized to the total electrode area. Reproduced with permission from ref 133. Copyright 2021 American Chemical Society.

The EIS test results confirmed that resistance of the cell was increased, which can be attributed to the formation of a secondary phase with low electrical conductivity. It was also noticed that, after 400 h of exposure, LSCF and the SDC layer agglomeration occurred.

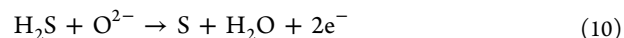
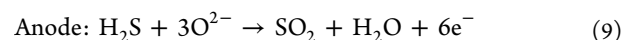
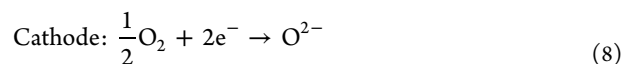
While the LSCF phase agglomeration occurs only due to particle coarsening, the SDC layer sintering can be enhanced due to Co and Fe inclusion effects, as they may serve as sintering agents. It was also noticed that microstructural changes were accelerated along with an increase in the cathodic current density, which may be attributed to an increase in the driving force for the formation of SrZrO_3 . This is accompanied by the change in the chemical potential distribution of Sr and Zr in each constituent layer due to an oxygen partial pressure drop.¹³⁸ The TEM analysis of the SDC/YSZ interface was conducted to detect elemental diffusion. After the 400 h of exposure time, YSZ, SrZrO_3 , and the SDC-YSZ solid solution and a Sm-rich YSZ-SDC solid solution phases were detected. It was noted that the electrical conductivity of the cell is low, which may also lead to performance deterioration. Therefore, it can be concluded that the cell degradation is caused by a combination of degradation factors, including LSCF and SDC agglomeration; triple-phase-boundary length reduction; SrZrO_3 formation; and a ceria-zirconia solid solution formation.

5. ROLE OF FEEDING GAS IMPURITIES ON THE PERFORMANCE OF LaFeO_3 -BASED SOECs

The formation of syngas by high temperature electrolysis needs a mixture of H_2O and CO_2 supplied to the cathodes surface.¹³⁹ As SOECs are a sustainable technology, a variety of different potential fuels have been considered. Each fuel has the potential to introduce different impurities that will vary depending on the fuel's initial composition, for instance, sulfur impurities, which are common for fossil fuels and biogas which could produce sulfur dioxide or hydrogen sulfide during the operation. H_2S is being widely reported as an activity inhibitor for Ni-YSZ cermet anodes in SOFC due to Ni surface poisoning.^{140,141} Several MIECs appeared to be sulfur tolerant^{142–144} and were exhaustively discussed by Gong et

al.,¹⁴⁵ in some cases, they also enhance the electrochemical activity.¹⁴⁶

Addo et al. tested the performance of $\text{La}_{0.3}\text{M}_{0.7}\text{Fe}_{0.7}\text{Cr}_{0.3}\text{O}_{3-\delta}$ ($\text{M} = \text{Sr}, \text{Ca}$) as the full cell anodes in the presence of 9 ppm of H_2S at 800 °C.¹⁴⁷ In the presence of H_2S , the polarization resistance dropped and was 4% smaller rather than in the initial gas composition (wet $\text{H}_2/\text{N}_2 = 30/70$). It was suggested that some adsorbed species like FeS might have formed, although no additional experiments were conducted to support this argument. Wenyi et al. have shown that perovskite structured material can be inactivated by the presence of H_2S .¹⁴⁸ $\text{La}_{0.75}\text{Sr}_{0.25}\text{Cr}_{0.5}\text{Mn}_{0.5}\text{O}_{3-\delta}$ (LSCM55) was tested as a potential material for the oxidation of H_2S in $\text{N}_2\text{--H}_2\text{S}$ (5%). The possible electrochemical reactions can be written as



In addition to electrochemical oxidation, it was proven that some chemical reactions take place on the surface with sulfur species involved. LSCM55 samples were heat treated at 800 °C in different atmospheres: (i) in 99.9% H_2 ; (ii) in 5% $\text{N}_2\text{--H}_2\text{S}$; and (iii) in 99.9% hydrogen switching to $\text{N}_2\text{--H}_2\text{S}$. XPS analysis presented on Figure 21 has shown that the dominating form of sulfur on the anode surface is sulfate (S^{6+}). In addition, XPS analysis of the O 1s region proved that lattice oxygen is predominantly used to form sulfate species. Moreover, the amount of sulfate formed decreases by reducing the oxygen content. This agrees with a study conducted by Wang et al. where they investigated the LSCF cathode performance in electrolysis mode for a half cell with dry air on the anode and dry air + 1 ppm of SO_2 on the cathode side.¹⁴⁹ The sulfate phase was found to be formed predominantly on electrochemically active sites, i.e., on the vacancies. The mechanism of sulfate formation was suggested as follows. SO_2 gas is adsorbed on the oxygen vacancy site. Then the gas molecule reacts with the lattice oxygen to form the SO_4^{2-} ion. Thus, by reducing the lattice oxygen concentration that process becomes unfavorable.

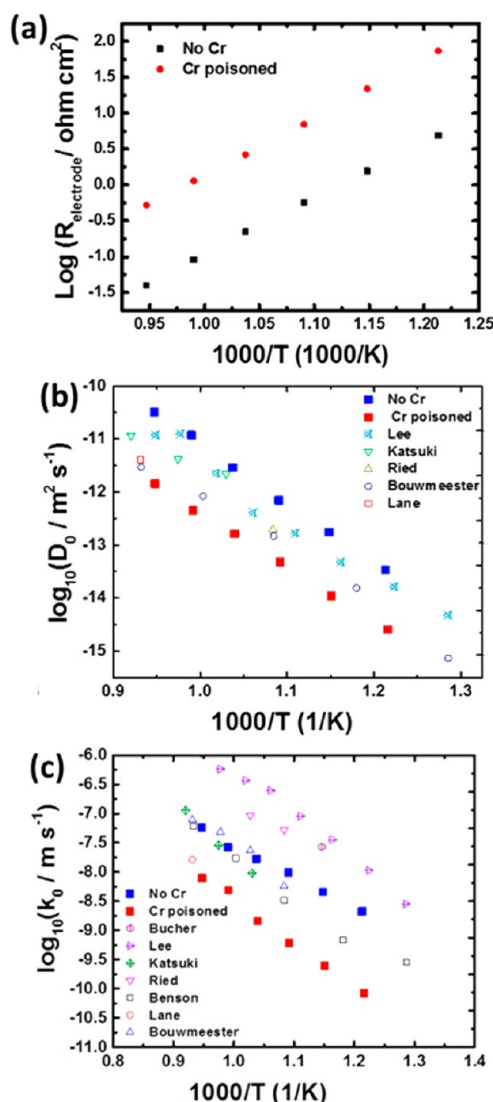


Figure 19. Arrhenius plots of the temperature dependence of (a) the cathode resistance $R_{\text{electrode}}$, (b) oxygen diffusion coefficient D_0 , and (c) Oxygen surface reaction equilibrium exchange rate constant k_0 . Literature data on D_0 and k_0 are shown for comparison. Reproduced with permission from ref 133. Copyright 2016 American Chemical Society.

In any case, strontium sulfate formation leads to a performance decrease, which may be attributed to the active site blocking due to Sr surface segregation. The same activity drop of LSCF was reported by Xie et al.¹⁵⁰ In addition, it was shown that the electrical conductivity of this material is also decreased by sulfur poisoning.

Another class of common impurities include the nitrogen oxides (NOx) that usually form when fossil fuels are burned at elevated temperature. As NOx severely contributes to the formation of smog and acid rain, some efforts were attempted to convert them to nitrogen in the SOC with a perovskite catalyst, such as LSM.^{59,151–153} None of these studies found anything to report regarding perovskite catalyst degradation. Another group of possible impurities includes the halogen-containing compounds. Predominantly, flue gas contains hydrogen chloride and hydrogen fluoride. To our best of knowledge, a few groups have tested the influence of flue gases with MIEC catalysts in SOCs. Li et al. has shown during a 1 h

test that no degradation was achieved in a hydrogen fuel cell with a HCl concentration of 8 ppm at 850 °C and 10 ppm at 750 °C.¹⁵⁴ Tremblay et al. reported a 17.4% loss of anode performance at 800 °C when a 20 ppm of HCl concentration was utilized.¹⁵⁷ The reaction between Ni and HCl to form NiCl_2 was suggested.¹⁵⁵ Hence, even low concentrations of HCl can significantly reduce the performance of the MIEC electrodes when metal nanoparticles are present on the surface. In addition, hydrogen halides are highly corrosive, especially at high temperatures. This may cause degradation of the metallic components of the supporting equipment and current collectors as well as ceramic materials that are used for sealing.^{156,157}

Aromatic hydrocarbons have been found to easily form coke on the catalyst active sites, which causes a drastic drop in the activity. This is likely caused by a slow reaction rate due to the presence of strong C–C bonds.¹⁵⁸ This was proven by many researchers studying cermet, such as Ni-YSZ or Cu-CeO₂.^{159–161} Nevertheless, this coke formation is predominantly reversible and can be removed after pure fuel is supplied.^{159,161} Lee et al. reported that during the diesel reforming reactions on the perovskite catalyst with compositions $\text{La}_{0.8}\text{Sr}_{0.2}\text{Cr}_{0.95}\text{Ru}_{0.05}\text{O}_3$ and $\text{Sm}_{0.8}\text{Ba}_{0.2}\text{Cr}_{0.95}\text{Ru}_{0.05}\text{O}_3$ coke formation can be observed.¹⁵⁸ It was proposed that the catalytic reaction is inhibited due to the occupation of active sites by carbon deposits. As coke formation is observed for both perovskite structured materials without cell polarization and a cermet catalyst, the same behavior could be expected for LaFeO_3 derived materials. There are many other compounds that can potentially be present in trace amounts such as AsH_3 , H_2Se , PH_3 , Sb , Hg , Cd , and Pb in the SOC feedstocks.¹⁶² Even though the quantity of these impurities is much smaller, their influence on the SOC performance is currently unknown and needs to be deeply investigated.

6. CONCLUSIONS

In summary, LaFeO_3 -based materials have the potential to replace cermet materials as electrodes in SOCs due to their high catalytic activity. Here, we reviewed the following key points about this material and its application in SOECs.

- Lanthanum ferrite-based materials crystallize in three distinct crystal structures based on temperature and composition, the orthorhombic phase is dominant at low temperatures and in low A-site doping systems, the rhombohedral phase evolves at higher temperatures or with heavily doped systems, and a cubic phase is usually only seen at high temperatures. The rhombohedral phase seems to be the preferred crystal structure due to LaFeO_3 catalytic activity at ~ 800 °C.
- The oxygen diffusion pathways in LaFeO_3 are important for understanding how the structure changes at the local scale when the material is operating as a catalyst at high temperatures. Several possible pathways have been computationally proposed using DFT studies.
- The catalytic performance of these materials could be improved by making composites or nanoparticle exsolution techniques. Both techniques have shown promise to improve the catalytic performance along with lowering the cost and making more durable cells.
- Lanthanum ferrite-based materials have some disadvantages such as poisoning by impurities in the feed gas. Several studies have shown that properties of LaFeO_3

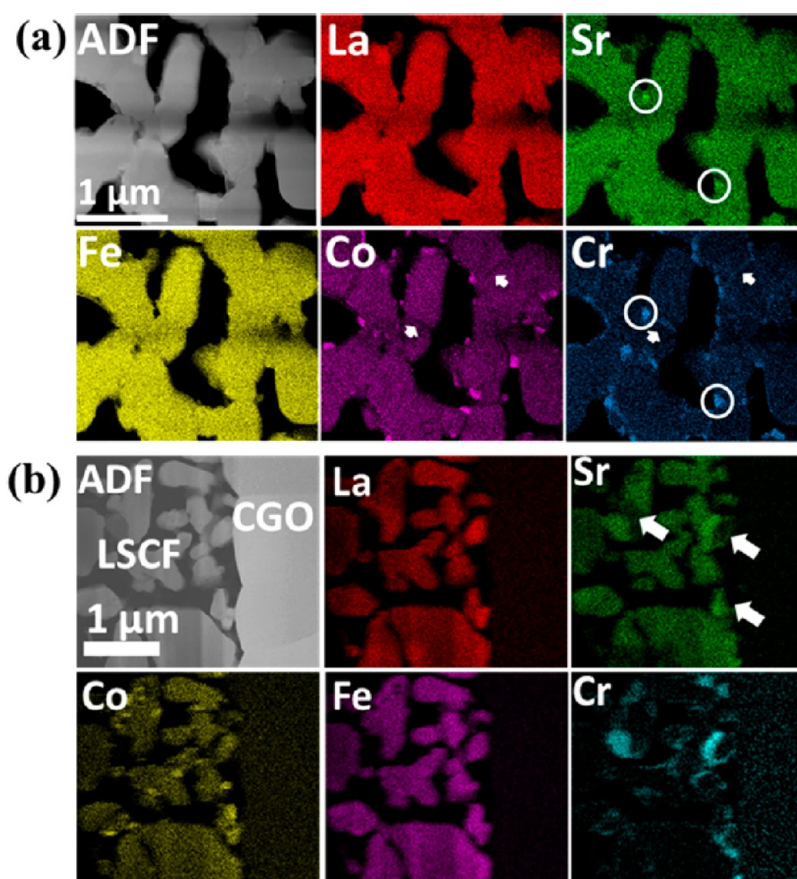


Figure 20. Representative STEM ADF-EDX elemental maps of the Cr poisoned sample obtained (a) from an area close to the LSCF6428 cathode surface and (b) from the LSCF6428/CGO interface. The intensity of the maps represents the net EDX counts after corrections for background and peak overlap. In (a), some SrCrO_4 grains are indicated with circles. The grain boundaries enriched with Cr and deficient in Co are denoted with arrows. In (b), the La rich and Sr deficient regions are also highlighted with arrows. Reproduced with permission from ref 133. Copyright 2016 American Chemical Society.

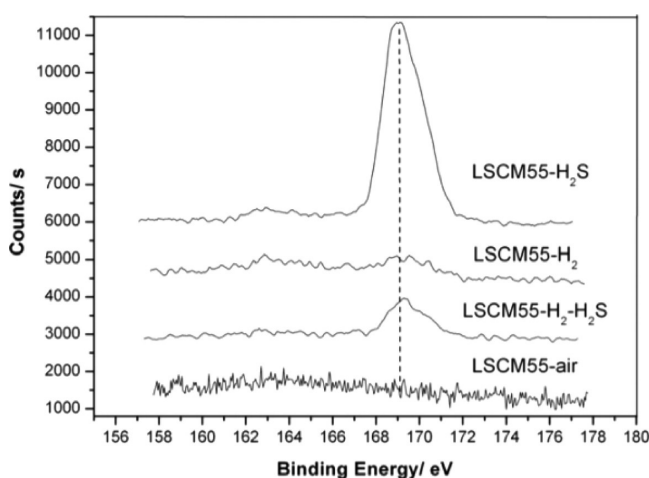


Figure 21. Representative XPS spectra of the S 2p region of anode LSCM55 after different treatments. Reprinted with permission from ref 148. Copyright 2012 Elsevier.

materials can be disrupted by H_2S , NO_x , and coking. This means that these materials may only be used with selective feed gas varieties or the cells need to be engineering to protect against these impurities.

As the field of SOECs develops, the usage of LaFeO_3 based materials will give us a better understanding of the material

requirements necessary to improve catalyst performance. So, a good understanding of the LaFeO_3 structure, diffusion pathways, and chemical stability are essential for further advancement of this technology.

AUTHOR INFORMATION

Corresponding Author

Venkataraman Thangadurai – Department of Chemistry, University of Calgary, Calgary, Alberta T2N 1N4, Canada; orcid.org/0000-0001-6256-6307; Email: vthangad@ucalgary.ca

Authors

Mykhailo Pidburtnyi – Department of Chemistry, University of Calgary, Calgary, Alberta T2N 1N4, Canada

Bryan Zanca – Department of Chemistry, University of Calgary, Calgary, Alberta T2N 1N4, Canada

Claude Coppex – Department of Chemistry, University of Calgary, Calgary, Alberta T2N 1N4, Canada

Santiago Jimenez-Villegas – Department of Chemistry, University of Calgary, Calgary, Alberta T2N 1N4, Canada;

orcid.org/0000-0001-8950-2899

Complete contact information is available at:

<https://pubs.acs.org/10.1021/acs.chemmater.1c00771>

Author Contributions

This article was written by graduate students for a class titled "CHEM 689.6 - Selected Topics in Physical Chemistry - Solid State Electrochemistry" offered by Dr. Venkataraman Thangadurai at the University of Calgary, Department of Chemistry in winter 2020. V.T. planned the review paper idea and mentored. B.Z. contributed to structural aspects, C.C. and S.J.V. contributed to the theoretical aspects, and M.P. wrote the electrochemistry and application aspects. All authors approved the paper.

Notes

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